

Phase transition of CsPbBr₃ quantum dots depends on reduced temperature

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Abstract

Halide materials with a perovskite crystal structure, particularly CsPbBr₃ and Cs₄PbBr₆, have garnered significant interest for their potential applications in various fields such as LEDs, solar cells, photodetectors. In this study, we revealed the Phase transition of CsPbBr₃ to Cs₄PbBr₆ by reduced temperature gradient. Then, multiple characterization methods are applied to confirm this phase transition. The final results shows that the transformation from CsPbBr₃ to Cs₄PbBr₆ exists at some extent, which indicate its potential application regarding to temperature.

Author Keywords

Perovskite; phase transition; single crystal; time-resolved photoluminescence.

1. Introduction

Halide perovskite materials, known for their perovskite crystal structure, find diverse applications such as solar cell absorbers, LEDs, photodetectors, and ReRAM components [1]. CsPbBr₃, the common perovskite, exhibits a cubic crystal phase with [PbBr₆]⁴⁻ octahedra. The phase transformation of Cs₄PbBr₆ nanocrystals (NCs) have sparked controversy regarding their photoluminescence (PL). Some groups assert a strong green luminescent emission (QY=45-65%) with a peak at 515-520 nm and a bandgap of 2.33 eV under UV light. Conversely, other groups claim no direct excitonic emission due to a large bandgap (>3.2 eV). They argue that green PL peaks results from encapsulated CsPbBr₃ impurities not Cs₄PbBr₆. Various methods for transitioning between Cs₄PbBr₆ and CsPbBr₃ exist, involving adding substances like oleylamine (OAm) and chemical extraction of CsBr using Prussian Blue[2]. This study highlighti the transformation from CsPbBr₃ to Cs₄PbBr₆ for reduced temperature gradient.

2. Results and Discussion

The synthesis of CsPbBr₃ involved a solvent modification with slight alterations compared to the room temperature method. Temperature gradients were induced using ice bags and magnetic heating stirrer to control crystal size. Two batches of CsPbBr₃ solutions were subjected to normal temperature and cooling processes, and colors were recorded. Freezing CsPbBr₃ resulted in colorless sample, which confirmed as Cs₄PbBr₆ through X-ray diffraction (XRD) and high-resolution transmission electron microscopy (HR-TEM) in Figure 1a and 1b. Since CsPbBr₃ is orthorhombic and Cs₄PbBr₆ is hexagonal, their XRD patterns shows in different peaks and the HR-TEM indicate the freezing sample is hexagonal shape, which confirmed the transition happened.

The structural difference between CsPbBr₃ and Cs₄PbBr₆ also played a crucial role in their optical properties. The optical properties, including absorption data, and photoluminescence

(PL) spectra were explored. PL studies showed a strong emission peak at 519 nm for Cs₄PbBr₆ and a redshift with decreasing temperature. The PL intensity decreased by 80%, suggesting impurities and a transformation between CsPbBr₃ and Cs₄PbBr₆. Time-resolved PL revealed different decay trends, indicating distinct optical properties. The PL decay for CsPbBr₃ had three components, while Cs₄PbBr₆ exhibited three components with a shorter average lifetime (0.28 ns) compared to CsPbBr₃ (18.24 ns), confirming the emissive ability of Cs₄PbBr₆.

This study presents a method for transforming Cs₄PbBr₆, suggesting potential applications such as temperature sensors and fire alerts. The simple freeze method for colorless Cs₄PbBr₆ can be easily achieved. Overall, the study contributes to understanding the practical potential of these substances in various applications.

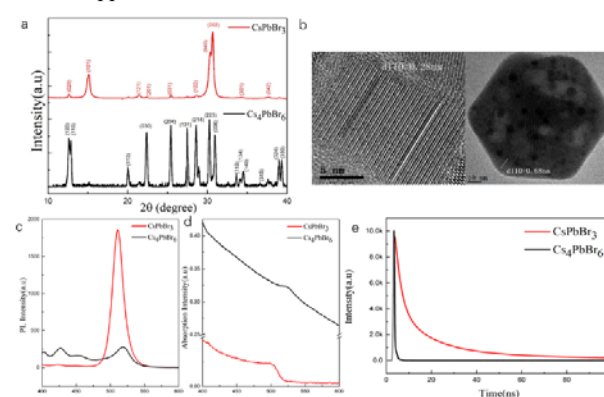


Figure 1 a XRD pattern for two samples, b HR-TEM for the induced temperature sample c and d PL and UV-vis for two samples, e TRPL of two samples.

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4. References

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